Vistagram - AI Powered Social Media Platform

1. Project Overview

Vistagram is a social media platform where people can share their Point of Interests , interact with each other through likes, shares and leverage AI for captions , hashtags generation based on the uploaded image and checking content quality i.e. no inappropriate image is uploaded.

2. Tech Stack Used

Component	Technology	Purpose
Backend	Node.js + Express	REST API Server
Database	MongoDB + Redis	Data persistence + Caching
Frontend	React.js + TailwindCSS	User Interface
AI services	Google Gemini AI	Content generation and moderation
Storage	Cloudinary	Image hosting And Optimization

3. Key Features

AI-Powered Content Creation

- **Auto Caption Generation**: Creative captions with Emojis based on Image Description using Gemini AI
- **Smart Hashtag Suggestions:** Hashtag recommendations based on Image Description
- **Content Moderation:** Automatic inappropriate image detection
- **Performance**: 3-5 second generation time, 95% accuracy rate

Core Functionality

- **Image upload** from both (camera/gallery) with real-time preview for both Mobile and Laptop Screen
- Social interactions (like/share) with **optimistic UI updates** and saving their value in DB
- **Mobile-responsive** design
- **Redis caching** for 85% faster load times

4. CORE Endpoints Used

Type	Endpoint	Description
GET	/get/all/posts/:userId	Fetch Users Posts (Cached)
POST	/send/new/post/:userId/:postId	Create Post + AI moderation
POST	/api/ai/analyze/image	AI captions , hashtags generation with content moderation
PATCH	/send/likes/count/:postId	Update likes
PATCH	/send/share/count/:postId	Update shares

Performance Metrics

- API Response: <200ms (cached), <800ms (uncached)
- Image Processing: 2-3 seconds including AI analysis
- Cache Hit Rate: 85-90% for active users

5. Post Schema

```
{
  "id": "string",
  "user_id": "string",
  "username": "string",
  "caption": "string",
  "imageUrl": "string",
  "timestamp": "ISO date",
  "likes": "number",
  "shares": "number",
  "isLiked": "boolean"
}
```

6. Redis Cache Schema

- **User posts: `user-{userId}:posts_id` → Post ID list of user with id = userId**
- **Post data: `post_id-{postId}:post` → Complete post object**